EECT/CE 6325 VLSI Design
Fall 2026

PROJECT #4: Cell Library
Due: Wed. Nov. 4

Project Introduction

For this project you will be creating your standard library of cells (without D-Flip Flop), which will be used for your final project. **NOTE: The poly width (length of the channel) must be 16nm for this project. No points for other sizes.**

Project Goals

1) You must design, layout and verify at least 10 different combinational standard cells. Note that the following are required:
   - INV
   - NAND2
   - NOR2
   - RIGHT - ENDCAP
   - LEFT - ENDCAP
2) It will be graded competitively, so the more cells you have and the more complex they are, the more points. However, be realistic with the time you have available!
3) Try not to have the same cells as another group; if you do, TAs will check more closely for copying.
4) All your cells should be placed side by side, next to each other, and have no DRC errors. All pins must be aligned horizontally with uniform spacing.
5) Generate a .lib file containing all the cells in your library using *PrimeLib*. To generate a .lib file, follow the instructions given in *PrimeLib by Synopsys for TSMCN16* document uploaded on eLearning.
6) Four fins for each of the pMOS and nMOS sections.
7) VDD is 0.8V & GND is 0V.
8) Generate abstract views of your cells, use instructions given in *Abstract View Generation* document uploaded on eLearning in Project 4 folder)

Project Rules & Requirements

1) All the cells should have the same height with VDD & GND rails aligned horizontally.
2) The input slew rate is 20 ps (0.2*Vdd to 0.8*Vdd and vice versa)
3) Use a 15 fF load capacitance when simulating.

What to Turn In (points are deducted for anything missing)

1) A cover page containing all the following information.
   - Name, NetID and project title
2) Each cell layout with **rulers** showing the dimensions of the cell. Part of grade depends on the clarity of your ***report***.
   - Show the distance between your pins
   - Show the length of the channel
   - Show the height & width of entire cell.
3) Also turn in the following for each cell in your library:
   - Simulation data that shows each transistor works in each cell
4) Please also hand in a layout showing all your cells lined up in a row with the boundary layers touching to demonstrate that they can be placed next to each other with no problems (yes, that means run DRC).
   - Show distance between pins to verify that they have a uniform pitch.
5) Upload the project report on eLearning.
6) Upload the zipped folder containing all layouts and schematics on eLearning

Grading Breakdown

Correct functionality of all cells	40%
Number and complexity of cells	20%
Correct pin spacing and sizing	20%
Report	20%